

AKASH SIKDAR SUDIP

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📍 298/5, Middle Pirerbag, Mirpur, Dhaka

EDUCATION

BSc in Industrial and Production Engineering

Jan' 2022 – Jun' 2026

- Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
- CGPA: 3.55 / 4.00

RESEARCH INTERESTS

Operations Research, Machine Learning & Data Analysis, Supply Chain Management, Process & System Optimization, Sustainable & Environmentally Responsible Operations

PUBLICATIONS

Conference Paper

Feb' 2025 – Dec' 2025

Debnath, A., Sudip, A. S., & Mandol, S. **Multi-Model Machine Learning Analysis of Ecological Load Ratio Using Classification, Regression, and Clustering Approaches in EU Countries.** *8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management, Dhaka, Bangladesh, December 20–21, 2025*

- Applied ensemble machine learning models (XGBoost, Gradient Boosting, AdaBoost, LightGBM) and Gaussian clustering algorithms for ecological load classification and forecasting.
- Achieved 98.67% classification accuracy and $R^2 = 0.9823$ for ELR forecasting, identifying major ecological drivers using feature importance analysis.

RESEARCH EXPERIENCE

Undergraduate Thesis Research (Under Review)

May' 2025 – Jun' 2026

Sudip, A. S., Debnath, A., Raihan, A., Bari, A. M. **Determinants of Environmental Sustainability in India: An ARDL Analysis Using the Load Capacity Factor.**

- Utilized ARDL, FMOLS, DOLS, and CCR econometric techniques with ADF/PP/DF-GLS stationarity tests to analyze environmental sustainability determinants in India (1990–2024).
- Found that a 1% rise in GDP per capita decreases LCF by 0.311% and confirmed robust long-run negative impacts of economic growth, fossil fuel energy production, and trade openness (ARDL $R^2 = 0.9928$).

Ongoing Research Work

Dec' 2025 – Current

- Developed interpretable ML models (Ridge, Lasso, Elastic Net, Bayesian Ridge, Quantile Regression) to predict Ecological Load Ratio and Ecological Deficit Per Capita across EU-27 countries, using a five-track hybrid feature selection framework with Borda count aggregation and VIF-based multicollinearity control.
- Applied SHAP, PDP, and LIME for model interpretability, identifying GDP per capita and freshwater resource availability as key drivers of ecological sustainability.

PROJECT EXPERIENCES

AquaPellet Feeder: Automated Aquaculture Feeding System

Mar' 2024 – May' 2025

Multidisciplinary Team Project, BUET

- Designed an automated aquaculture feeding system targeting improved pellet distribution efficiency and reduced feed wastage in aquaculture operations.
- Applied process flow analysis and CAD-based mechanical design; investigated sustainability outcomes through feed utilization modelling.

IoT-Based Fire Detection System for Garments Industry

Jul' 2025

Industrial Safety Research Project BUET

- Designed and built a real-time IoT fire detection and warning system addressing critical safety gaps in Bangladesh's Ready-Made Garments (RMG) sector.
- Integrated sensor networks for multi-point monitoring with instant alert dispatch. Applied industrial safety standards and process control principles.

Humanoid Robot Exterior Design

Feb'2025

CAD/CAM Sessional

- Course Designed and assembled individual components of a humanoid robot exterior in SolidWorks, producing detailed part drawings and a fully constrained assembly.
- Rendered the final assembly using SolidWorks Visualize, demonstrating proficiency in parametric modelling, part-to-assembly workflow, and engineering aesthetics.

PROFESSIONAL EXPERIENCE

Industrial Attachment — British American Tobacco Bangladesh

Nov – Dec' 2025

Savar Manufacturing Plant, Dhaka, Bangladesh

- Authored 25+ Standard Operating Procedures (SOPs) for 10+ utility assets, applying technical writing and process analysis skills to systematize operational knowledge.
- Documented safety compliance protocols for use by 100+ plant technicians, gaining direct exposure to large-scale industrial process management.

SKILLS & COMPETENCIES

Programming: Python, MATLAB, C

Design Software: AutoCAD, SolidWorks, CATIA, Ansys

Engineering & Technical: Ergonomics, Lean Manufacturing, Six Sigma (Basics), Quality Control

Data Analysis: Microsoft Excel, Statistical Analysis, Data Visualization

AWARDS & HONOURS

Dean's List Award from Faculty of Mechanical Engineering, BUET

Jun'2022

University Merit from BUET

Jun'2022

Government Board Scholarship, Jashore Board, Bangladesh

2018, 2016

EXTRACURRICULAR ACTIVITIES

Vice President — BADHAN, BUET Zone

Apr'2025 – May' 2026

- Coordinated voluntary blood donation activities; managed collection of 300+ units of blood for patients in need.

General Secretary — Saraswati Puja Organizing Committee, BUET

Jan' 2025

- Hosted the Ambassador of Nepal to Bangladesh as an honored guest at an event attended by 5,000+ participants in Dhaka.

Logistics Secretary — AIPE, BUET

May' 2025 – May' 2026

- Organized and oversaw logistics for departmental academic and professional programs.

Student Magazine Editor

Jan' 2026

- Edited and supervised content for three student magazines, ensuring publication quality and adherence to deadlines.

REFERENCES

Dr. Nikhil Ranjan Dhar

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